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ACADEMIC VAULT

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ANA 101: Human Anatomy I - Cells, Tissues and the Skeletal System

Learn anatomy by systems and relationships rather than by isolated lists

FACULTY PRIMER

Faculty Primer

Faculty of Basic Medical Sciences
Human Anatomy | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Anatomy has a reputation for impossible memorisation. It becomes manageable when you learn structures by relationship, function and clinical relevance. This pack covers anatomical terminology, tissue types, bone structure and the axial and appendicular skeleton, with a revision method built around drawing and recall.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Use correct anatomical position, planes, directional terms and movement terminology.
- Describe the four basic tissue types and identify their functions.
- Explain bone structure, growth and ossification, and classify bones by shape.
- Name and locate the major bones of the axial and appendicular skeleton.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The language of anatomy

Precise terminology prevents dangerous ambiguity. Learn it before you learn structures.

- Anatomical position: standing upright, palms facing forward, feet together.
- Planes: sagittal (left and right), coronal (front and back), transverse (upper and lower).
- Directional terms: superior and inferior, anterior and posterior, medial and lateral, proximal and distal, superficial and deep.
- Movements: flexion, extension, abduction, adduction, rotation, circumduction, pronation and supination.
- Body cavities: cranial, vertebral, thoracic, abdominal and pelvic.

- Regions: use the correct regional term, for example brachial for arm and femoral for thigh.

> **Name the relationship, not just the part**
> In practicals you will be asked where a structure lies relative to another. Practise saying 'the median nerve passes medial to the brachial artery' rather than only labelling a diagram.

The four basic tissues

Organs are combinations of four tissue types, each specialised for a job.

- Epithelial tissue: covers and lines surfaces; classified by cell shape and layer number.
- Connective tissue: supports, binds and protects; includes bone, cartilage, adipose and blood.
- Muscle tissue: skeletal (voluntary, striated), cardiac (involuntary, intercalated discs), smooth (involuntary, non-striated).
- Nervous tissue: neurons transmit impulses; neuroglia support and protect.
- Histology requires practice at identifying tissue from appearance, not from written descriptions alone.
- Always relate structure to function: thin squamous epithelium suits diffusion; stratified epithelium suits protection.

Bone and the skeleton

Bone is living tissue that remodels throughout life, responding to load and hormonal signals.

- Long bone structure: diaphysis, epiphyses, periosteum, medullary cavity, articular cartilage.
- Compact bone forms the outer shell; spongy bone forms the inner lattice.
- Ossification: intramembranous for flat bones of the skull; endochondral for most long bones.
- Bone cells: osteoblasts build, osteocytes maintain and osteoclasts resorb.
- Axial skeleton: skull, vertebral column, ribs and sternum.
- Appendicular skeleton: pectoral girdle, upper limbs, pelvic girdle and lower limbs.

> **Draw from memory daily**
> Ten minutes of blank-page drawing beats an hour of highlighting. Draw, label, check, correct, repeat the next day.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Define anatomical position.

Answer: Standing erect, face forward, upper limbs by the sides with palms facing forward, and feet together.

2. Which tissue type lines the inner surface of blood vessels?

Answer: Simple squamous epithelium, called endothelium, suited to diffusion and a smooth flow surface.

3. Distinguish intramembranous from endochondral ossification.

Answer: Intramembranous ossification forms bone directly from mesenchyme, as in skull flat bones. Endochondral ossification replaces a cartilage model, as in most long bones.

4. Name the bone cells responsible for bone resorption.

Answer: Osteoclasts.

5. Which plane divides the body into equal left and right halves?

Answer: The midsagittal plane.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write the anatomical position and all directional terms from memory.
- [ ] Draw and label a long bone with six structures.
- [ ] Identify the four tissue types from microscope images.
- [ ] Name the bones of the axial skeleton without notes.
- [ ] Label the bones of the upper and lower limbs on a blank diagram.
- [ ] Explain the roles of osteoblasts, osteocytes and osteoclasts.
- [ ] Practise ten spotting questions with a study partner.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: